

Matrix-Residual Graph Neural Networks for Graph Classification

Jingchao Wang, Xiaodan Cui*, and Pengming Hu

School of Computer Science

Zhongyuan University of Technology

Zhengzhou 450007, China

static@zut.edu.cn, cuixiaodan2026@163.com, hupengming@zut.edu.cn

*Corresponding author

Abstract—Residual connections are widely used to stabilize graph neural networks, yet residual design is usually treated as a one-dimensional depth-wise choice. Multi-branch graph classifiers open a richer design space in which residual information can flow along depth, across branches, or through a local two-dimensional branch-by-layer neighborhood. This paper studies Matrix-Residual Graph Neural Networks as a controlled residual-topology framework for graph classification. The model family represents hidden states on a branch-by-layer grid and compares five information-flow designs: plain message passing, vertical residual reuse, horizontal branch-wise exchange, local matrix residual reuse, and sparse or gated matrix reuse. The main benchmark evaluates six datasets, four message-passing operators, and five graph-level folds under a shared training protocol, alongside branch-count ablations, a mechanism analysis, and a controlled synthetic benchmark spanning five graph families and two to eight classes. No single residual topology is universally optimal across datasets and operators, and branch-count ablations reveal dataset-dependent operating regions rather than monotonic capacity gains. The synthetic benchmark shows that simple baselines remain competitive on low-class tasks, whereas matrix residual reuse becomes consistently more favorable as structural class granularity increases. Residual topology, branch count, and residual filtering should therefore be selected jointly; matrix-style reuse is best understood as a tunable residual family rather than a universal replacement for simpler vertical or horizontal residual paths.

Index Terms—graph classification, graph neural networks, residual connections, multi-branch architectures, matrix residual reuse

I. INTRODUCTION

Graph neural networks (GNNs) learn graph representations by repeatedly aggregating information from local neighborhoods [1]. This message-passing view has become a standard tool for graph classification because it combines node attributes, local connectivity, and graph-level structure in a single differentiable model [2], [3]. Graph classifiers are used well beyond the standard benchmarks: to identify plant vacuole proteins from contact maps [4], to propagate functional annotation through plant knowledge graphs [5], and for knowledge graph completion, where recent work enriches node representations with multi-granularity contextual semantics [6], global and

local context [7], and fused textual evidence [8]. In such tasks, graph labels may depend on both local motifs and global organization, so useful intermediate evidence must survive until graph-level pooling [9].

Increasing depth or adding parallel branches expands capacity but also changes information transport. Deep GNNs may suffer from over-smoothing, where node representations become too similar across propagation steps [10], or from over-squashing, where long-range information is compressed into limited-size messages [11], [12]. Both are critical for graph-level prediction, because the final readout depends on the intermediate representations that reach the pooling stage. Residual connections are a common response [13], [14], as is DropEdge [15]. Yet residual connectivity is usually treated as a default layer-wise component rather than as the object of study: most residual graph classifiers reuse information along depth within a single stream.

Once a model contains multiple branches, depth-wise reuse is only one possible route. Residual information may also move across neighboring branches at the same depth, or through diagonal branch-depth neighborhoods. Prior work has strengthened message-passing operators, pooling, and higher-order neighborhood mixing [16], [17], but these designs do not isolate the residual-routing question: given several branch states at a given depth, should the model reuse only the previous state of the same branch, exchange states laterally, or combine both in a local two-dimensional neighborhood?

This paper studies residual reuse as a structured design problem on a branch-by-layer grid. The question is not whether skip connections should exist, but where residual signals should flow. The grid exposes three controlled neighborhoods: vertical reuse along depth, horizontal exchange across branches, and matrix-style reuse combining same-branch, neighboring-branch, and diagonal paths. A sparse/gated variant tests whether matrix residual traffic should be filtered before injection. The study is deliberately controlled: the classifier, readout, fold protocol, and backbone options are held fixed, so that only the residual topology and its control mechanism vary. Recent models may reach stronger absolute performance by changing several components at once [18]; here those components are intentionally fixed so that residual topology remains identifiable, following the same discipline as fair-comparison

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protocols for graph classification [19], [20]. Because real benchmarks span a limited range of structural class granularity, we complement them with controlled synthetic graph families whose labels are generated by structural parameters.

The work addresses three research questions. **RQ1:** Does residual topology affect performance under a shared graph-classification protocol? **RQ2:** How do branch count, residual traffic, and mechanisms explain performance changes? **RQ3:** When is matrix-style residual reuse more effective on real data and on controlled synthetic multi-class structures?

The contributions are:

- **A branch-by-layer formulation.** Vertical, horizontal, matrix, and sparse/gated matrix residual families are expressed as local neighborhoods on the same grid, making their information-flow differences explicit.
- **A controlled real-data and synthetic benchmark.** Classifier, readout, fold protocol, and backbone choices are matched across all comparisons, and a structural stress test spans five controlled graph families and two to eight classes.
- **A mechanism analysis with conservative rerun checks.** Accuracy is analyzed alongside branch diversity, branch cosine similarity, centered kernel alignment, gradient norms, and residual activity. Sensitivity scans only identify candidates; selected settings are rerun across folds before being treated as evidence.

II. RELATED WORK

Message passing. Modern graph learning builds on the spectral convolution of Kipf and Welling [1]. GraphSAGE introduced inductive neighborhood aggregation [21], graph attention networks weighted neighbors by learned attention [22], and the message-passing framework unified these designs for molecular property prediction [2]. The Weisfeiler–Lehman connection established by GIN shows that injective multiset aggregation is central to the discriminative power of this family [23], and surveys document the resulting design space [3].

Deep GNNs and residual connectivity. Depth is not free: over-smoothing makes node representations converge as propagation depth grows [10], while over-squashing compresses long-range information into fixed-size messages [11], [12]. DropEdge randomly removes edges during training [15], and residual or dense skip paths carry earlier states forward [13]. Residual gated graph convolutional networks learn how strongly historical states should be reinjected [14]. These approaches reuse depth-wise state within a single stream; our work keeps the same reuse principle but asks which local residual neighborhoods on a branch-by-layer grid should be connected.

Multi-branch and higher-order architectures. Higher-order neighborhood mixing improves over single-hop aggregation [16]. Hierarchical pooling adds a second axis of structure by coarsening graphs during processing [9], and a recent survey catalogues these designs [17]. Such work increases the expressive or aggregational capacity of the propagation layer. In contrast, we hold the propagation operator fixed and vary

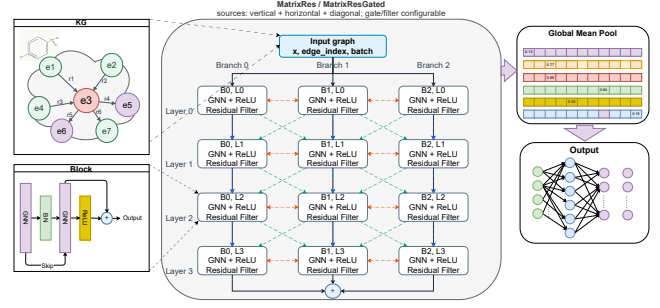


Fig. 1. Compared residual neighborhoods on the branch-by-layer grid; the backbone and graph-level readout are held fixed while the residual routes change.

only how hidden states are reused, so that the residual topology itself becomes the identifiable factor.

III. METHOD

A. Notation and Shared Pipeline

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a graph with node feature matrix $\mathbf{X} \in \mathbb{R}^{n \times d_{\text{in}}}$ and adjacency structure $\mathbf{A}$. Given a dataset $\mathcal{D} = \{(\mathcal{G}_i, y_i)\}_{i=1}^N$, the objective is to learn $f_\theta : \mathcal{G} \rightarrow \hat{y}$, where $\hat{y}$ is the predicted graph label. Let B be the number of branches and L the number of message-passing layers, and write the hidden state at branch b and layer ℓ as $\mathbf{H}_b^{(\ell)}$, with $b \in \{1, \dots, B\}$. A branch-local propagation step is

$$\mathbf{Z}_b^{(\ell)} = \phi_b^{(\ell)} \left(\mathbf{H}_b^{(\ell-1)}, \mathbf{A} \right), \quad (1)$$

where ϕ is one of four standard operators: GCNConv [1], GATConv [22], SAGEConv [21], or GINConv [23]. The residual-enhanced update is

$$\mathbf{H}_b^{(\ell)} = \mathbf{Z}_b^{(\ell)} + \sum_{(b', \ell') \in \mathcal{N}(b, \ell)} \mathcal{R}_{(b', \ell') \rightarrow (b, \ell)} \left(\mathbf{H}_{b'}^{(\ell')} \right). \quad (2)$$

Only the residual neighborhood $\mathcal{N}(b, \ell)$ and the residual transform $\mathcal{R}$ change across architectures. After the final layer, each branch is pooled to a graph-level vector by global mean pooling, and the branch vectors are combined before classification with a cross-entropy loss. We deliberately avoid architecture-specific pooling heads such as differentiable pooling [9], because the goal is to isolate residual topology rather than to compare readout mechanisms. Fig. 1 summarizes the shared pipeline and the compared residual routes. The four operators were chosen as canonical representatives of spectral-style convolution, attention-based aggregation, neighborhood sampling and aggregation, and injective multiset aggregation.

B. Residual Neighborhoods

Plain has no residual neighborhood, $\mathcal{N}(b, \ell) = \emptyset$. **Vertical-Res** injects the previous layer of the same branch, $\mathcal{N}(b, \ell) = \{(b, \ell - 1)\}$. **HorizontalRes** exchanges information across adjacent branches at the same depth, $\mathcal{N}(b, \ell) = \{(b-1, \ell), (b+1, \ell)\}$, with boundary branches using only valid neighbors; when $B = 1$ this neighborhood is empty. **MatrixRes** combines vertical, horizontal, and diagonal neighbors:

$$\mathcal{N}(b, \ell) = \{(b, \ell-1), (b-1, \ell), (b+1, \ell), (b-1, \ell-1), (b+1, \ell-1)\}. \quad (3)$$

This local stencil is intentionally not dense all-to-all mixing: it preserves an interpretable residual structure over the branch-by-layer grid. Invalid boundary neighbors are removed, so when $B = 1$, MatrixRes keeps the same-branch vertical path after the first layer but has no branch-neighbor routes.

MatrixResGated uses the same neighborhood but controls the injected signal through the residual transform:

$$\mathcal{R}(\mathbf{U}) = \begin{cases} \alpha \mathbf{U}, & \text{identity filtering with a gate,} \\ \alpha \text{softshrink}_{\lambda}(\mathbf{U}), & \text{sparse filtering with a gate.} \end{cases} \quad (4)$$

Plain, VerticalRes, HorizontalRes, and MatrixRes use identity residual filtering. MatrixResGated uses sparse filtering with $\lambda = 0.05$ and a learnable scalar gate initialized at 0.8; when $B = 1$ it still applies gated sparse filtering to the remaining matrix-residual paths. Top- k filtering and fixed gates are supported but are not used for the reported results.

C. Experimental Setup

The main benchmark uses six TUDataset collections [20] accessed through PyTorch Geometric [24]: PROTEINS, DD, ENZYMES, MUTAG, AIDS, and Mutagenicity. PROTEINS and DD are binary protein-structure tasks, ENZYMES is six-class, and MUTAG, AIDS, and Mutagenicity provide molecular-graph robustness checks. Graphs are loaded with their provided features; no handcrafted descriptors, augmentation, or edge rewiring are used, so that the comparison reflects residual topology rather than feature engineering.

All results use graph-level five-fold evaluation. For fold k , test accuracy is $\text{Acc}_k = |\mathcal{T}_k|^{-1} \sum_{(\mathcal{G}_i, y_i) \in \mathcal{T}_k} \mathbf{1}[\hat{y}_i = y_i]$, and we report the mean over folds together with the fold-level standard deviation. Accuracy is the primary metric because every benchmark has discrete graph labels. For the synthetic benchmark we additionally report macro-F1, which weights each class equally, and chance-normalized accuracy, $\text{NAcc} = (\text{Acc} - 1/C)/(1 - 1/C)$, because raw accuracy is not comparable across class counts. Mechanism metrics such as branch diversity, cosine similarity, centered kernel alignment (CKA), residual activity, and gradient norm are used as diagnostics, not as optimization targets.

To test residual topology under controlled structural granularity we add a synthetic benchmark over five graph families: Bernoulli random graphs (ER), Barabási–Albert (BA) graphs, stochastic block models (SBM) [25], Watts–Strogatz (WS) graphs [26], and a regular-degree control. For each family we generate seven datasets with $C \in \{2, \dots, 8\}$ classes, 200 graphs per class, and 30–80 nodes per graph. Node features are eight-dimensional vectors derived from normalized degree and random channels; they do not encode the label. Labels are controlled by family-specific structural parameters: edge probability for ER [27], preferential-attachment count for BA [28], within- and between-block probabilities for SBM, rewiring probability for WS, and fixed degree for the regular control. These families serve as stress tests, not as replacements for real benchmarks.

TABLE I
KEY EXPERIMENTAL SETTINGS

Parameter	Value
Operators	GCNConv, GATConv, SAGEConv, GINConv
Branches / layers	$B = 3$; $L = 4$ (DD: $L = 3$)
Hidden dimension	64
Evaluation	Five-fold graph classification
Plain/Vertical/Horizontal	200 epochs, patience 60, lr 0.005, wd 10^{-4} , dropout 0.5
MatrixRes/MatrixResGated	240 epochs, patience 80, lr 0.003, wd 5×10^{-5} , dropout 0.3
Schedule	ReduceLROnPlateau, factor 0.5, patience 15, min lr 10^{-5}
Gradient clipping	Norm 2.0

TABLE II
WINNER COUNTS AND MEAN RANKS OVER THE 24 DATASET–OPERATOR COMBINATIONS

Model	Wins	Mean rank	GCNConv wins
Plain	5	3.04	PROTEINS
VerticalRes	2	3.38	—
HorizontalRes	3	3.54	DD
MatrixRes	9	2.12	MUTAG, AIDS, Mutagenicity
MatrixResGated	5	2.92	ENZYMES

Table I lists the shared settings. Models are optimized with Adam [29], and the checkpoint with the lowest validation loss is retained for test evaluation. Branch-count ablations evaluate $B = 1, \dots, 8$ for HorizontalRes, MatrixRes, and MatrixResGated on PROTEINS and DD under GCNConv. Sensitivity scans run on fold 0 at $B = 3$, varying learning rate, dropout, hidden dimension, sparsity strength, and gate initialization; selected candidates are then rerun across all five folds. Experiments ran on an Ubuntu server with an AMD Ryzen 7 5700X CPU and an NVIDIA RTX 3090 GPU.

IV. RESULTS

All results are five-fold mean best test accuracy $\pm$ fold-level standard deviation unless stated otherwise.

A. Overall Benchmark

The main benchmark asks whether residual topology has a stable winner across datasets and operators. It does not. Across the 24 dataset–operator combinations (Table II), MatrixRes is the most frequent winner with 9 wins and the best mean rank (2.12); MatrixResGated and Plain win 5 combinations each, HorizontalRes wins 3, and VerticalRes wins 2. Matrix-style reuse therefore has the most favorable aggregate profile, but every topology is strongest in some setting. The per-dataset winners are mixed: the matrix family dominates MUTAG, AIDS, and Mutagenicity, whereas PROTEINS alternates between Plain and HorizontalRes and DD alternates among HorizontalRes, MatrixRes, and VerticalRes. ENZYMES remains difficult and shows no stable preference.

ENZYMES should be read cautiously: excluding it leaves MatrixRes with 9 wins over 20 combinations and a mean rank of 2.05, so the aggregate preference is unchanged.

To remove variation caused by the operator, we also report the GCNConv slice separately (Table III, Fig. 2). The same conclusion holds: MatrixRes is the strongest choice

TABLE III
GCNCONV BENCHMARK AT $B = 3$ (MEAN $\pm$ STD. OVER FIVE FOLDS)

Dataset	Plain	Vertical	Horizontal	Matrix	MatrixGated
PROTEINS	0.7044	0.6993	0.6992	0.6973	0.6886
DD	0.7053	0.7143	0.7181	0.7127	0.7141
ENZYMES	0.2683	0.2550	0.2633	0.2750	0.2933
MUTAG	0.7556	0.7451	0.7343	0.7609	0.7556
AIDS	0.8215	0.8210	0.8130	0.8365	0.8150
Mutagenicity	0.7644	0.7667	0.7738	0.7784	0.7701

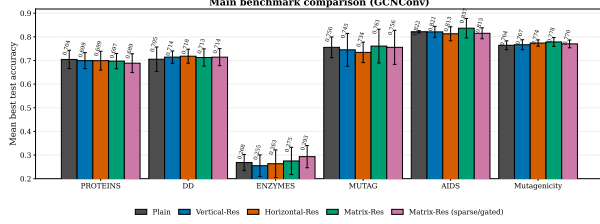


Fig. 2. GCNConv benchmark at $B = 3$; fold-level standard deviations range from 0.005 to 0.072.

TABLE IV
SYNTHETIC SCALING AVERAGED OVER FIVE GRAPH FAMILIES AND FOUR OPERATORS

Classes	Best model	Plain	Matrix	MatrixGated	Best NAcc
2	Plain	0.9585	0.9481	0.9578	0.9170
3	HorizontalRes	0.8774	0.8600	0.8772	0.8200
4	MatrixResGated	0.8002	0.8083	0.8111	0.7481
5	MatrixResGated	0.7233	0.7280	0.7296	0.6621
6	MatrixResGated	0.6719	0.6827	0.6878	0.6254
7	VerticalRes	0.6314	0.6429	0.6395	0.5945
8	MatrixResGated	0.5865	0.6002	0.6035	0.5469

on MUTAG, AIDS, and Mutagenicity, while simpler residual paths remain preferable on PROTEINS, DD, and ENZYMES. Residual topology matters, but its value depends on the dataset.

B. Synthetic Multi-Class Scaling

The synthetic benchmark asks whether residual topology behaves differently as structural class granularity increases under controlled graph-generation rules. Table IV summarizes the trend averaged over the five families and four operators. Low-class settings do not favour matrix-style reuse: Plain is best at $C = 2$ and HorizontalRes at $C = 3$. From $C = 4$ onward matrix-style reuse becomes more competitive, and MatrixResGated is the strongest mean-accuracy model at $C = 4$, $C = 5$, $C = 6$, and $C = 8$. Over the higher-class regime $C = 4, \dots, 8$, MatrixResGated reaches 0.6943 accuracy and 0.6324 normalized accuracy against 0.6827 and 0.6183 for Plain, while MatrixRes has the best macro-F1 (0.6880). MatrixRes also degrades least from $C = 2$ to $C = 8$ (0.3479 points, against 0.3720 for Plain), suggesting that dense local matrix reuse is relatively stable as the label space becomes finer.

At the family level, MatrixRes is strongest in the higher-class BA, ER, and regular-degree settings, where labels are tied to degree growth, edge probability, or degree-uniform structural change. These families share a pattern: the label space is formed by related structural regimes rather than unrelated categories, so local matrix reuse can combine depth-

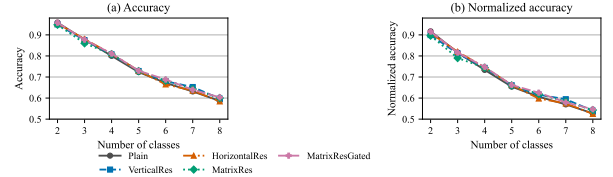


Fig. 3. Synthetic scaling from two to eight classes, averaged over five graph families and four operators.

TABLE V
BRANCH-COUNT ABLATION: BEST AND WORST SETTINGS

Dataset	Model	Best B	Best acc.	Worst B	Worst acc.
PROTEINS	HorizontalRes	4	0.7125	1	0.6801
PROTEINS	MatrixRes	6	0.7062	1	0.6909
PROTEINS	MatrixResGated	3	0.7098	4	0.6711
DD	HorizontalRes	2	0.7275	4	0.7071
DD	MatrixRes	2	0.7206	7	0.7071
DD	MatrixResGated	1	0.7283	7	0.7053

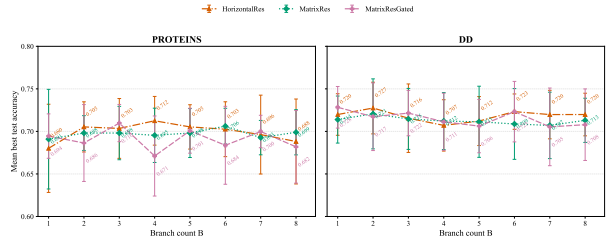


Fig. 4. Branch-count ablation on PROTEINS and DD under GCNConv. Accuracy rises only within an effective operating region before saturating or declining.

wise refinement with neighboring-branch exchange. SBM is community-driven and less favourable to matrix-style reuse, though MatrixRes remains close to the best result.

C. Branch-Count Ablation and Mechanism Analysis

Branch count is not merely a capacity parameter, because it also changes residual-path density. Table V reports the best and worst branch counts per topology. On PROTEINS the useful region is moderate (HorizontalRes peaks at $B = 4$, MatrixRes at $B = 6$, MatrixResGated at $B = 3$); on DD it is smaller (HorizontalRes and MatrixRes peak at $B = 2$, MatrixResGated is strongest already at $B = 1$). Branch expansion is therefore not monotonic and must be matched to task structure.

The mechanism metrics explain this split. On PROTEINS, HorizontalRes improves from $B = 1$ to $B = 4$ while branch diversity rises from 0.0000 to 1.1450 and branch cosine falls from 1.0000 to 0.8417, but diversity alone is not sufficient: at $B = 8$ diversity increases further to 1.5441 while accuracy drops to 0.6881. MatrixRes follows the same pattern at a larger scale, peaking at $B = 6$ with a diversity of 3.7755. On DD the best settings have nearly saturated branch cosine (0.9969 for HorizontalRes and 0.9991 for MatrixRes at $B = 2$) and CKA of 1.0000, so extra residual traffic is unlikely to create new functional separation; larger branch counts raise diversity without improving accuracy. MatrixResGated behaves as genuine residual control rather than a dense path: its active

ratio is 0.3121 at the best PROTEINS setting and 0.2352 at the best DD setting, while the learned gate stays near 0.78–0.84 rather than collapsing. These are diagnostic correlations rather than causal interventions. The gated variant does not dominate MatrixRes overall, so sparse filtering is best treated as a traffic-control mechanism, not a universal improvement.

D. Sensitivity and Five-Fold Validation

Fold-0 scans only identify candidates. On PROTEINS, MatrixRes does not improve over its fold-0 baseline in the tested sweep, while MatrixResGated has a mild-sparsity candidate ($\lambda = 0.02$ raises fold-0 accuracy from 0.6726 to 0.6951; stronger sparsification is harmful). On DD both models prefer lighter hidden dimensions, reaching 0.7595 and 0.7511 at $d = 32$. Rerunning selected candidates across all five folds prevents over-claiming: on DD the $d = 32$ MatrixResGated candidate stays competitive at 0.7215 ± 0.0330 while cutting parameters from 42,371 to 15,043, whereas on PROTEINS the sparsity candidate does not preserve its fold-0 advantage (0.6909 ± 0.0447) and remains unstable rather than a confirmed gain.

V. DISCUSSION

A. RQ1: Residual Topology and Performance

The main benchmark argues against a universal residual topology: MatrixRes is the most frequent winner with the best mean rank, but all five model families win at least two of the 24 dataset–operator combinations, and the GCNConv slice also has dataset-dependent winners. The useful residual neighborhood therefore depends on dataset-specific tolerance for branch specialization, residual traffic, and optimization complexity. This is consistent with the broader literature, where changing the propagation operator, normalization, or pooling design shifts the preferred architecture [9], [21], [23]. The present study adds residual topology itself as another source of that shift. Matrix-style reuse can be read as a local residual stencil over the branch-by-layer grid, analogous in spirit to structured neighborhood mixing [16] but applied to hidden-state reuse rather than graph adjacency; its advantage is architectural organization, not a new message-passing primitive.

B. RQ2: Branch Count and Mechanisms

Increasing B adds branches, residual paths, and optimization interactions at once, so it is not a simple capacity knob. PROTEINS tolerates a moderate budget (peaks at $B = 4$ –6) whereas DD prefers $B = 1$ –2. The mechanism metrics are consistent with this split: on PROTEINS the best settings combine increased branch diversity with reduced branch cosine, while on DD the best settings have nearly saturated cosine and CKA, so additional residual traffic is unlikely to create new functions. Diversity may rise with B but does not by itself improve classification when branch similarity stays high or gradients weaken. This rise-then-fall behaviour is a practical warning for residual GNN design, since residual connections are often introduced to make deeper or wider models easier to

optimize [13]. The balance between functional diversity and residual interference differs sharply between two datasets that are both routinely treated as protein-structure benchmarks.

C. RQ3: When Matrix-Style Reuse Helps

Sparse/gated reuse is not uniformly better than dense matrix reuse: MatrixRes wins more combinations overall. MatrixRes-Gated remains useful as a controlled way to suppress weak residual coefficients, and the mechanism table confirms that it keeps fewer active pathways than dense MatrixRes while the gate stays at a moderate value. Its value is not that sparsity always improves accuracy, but that filtering helps when matrix reuse would otherwise inject too much low-value traffic. This places it between two familiar ideas: like residual gated graph networks it learns how strongly historical information should be injected [14], and like sparsification methods it reduces the influence of weak pathways [15]. The implementation is deliberately simple—a scalar gate and soft-shrink filtering rather than a large routing network—which keeps the ablation interpretable. Candidate settings must still be validated across folds: the DD hidden-dimension candidate survives that check and yields a smaller, more efficient model, whereas the PROTEINS sparsity candidate does not and should not be claimed as a gain.

The synthetic benchmark separates class granularity from real-dataset idiosyncrasies. Simple residual choices remain competitive at two and three classes, which guards against the overbroad claim that matrix-style reuse is always needed. From four classes onward the pattern reverses: MatrixRes-Gated has the best average and normalized accuracy, MatrixRes has the best macro-F1 and the smallest degradation from $C = 2$ to $C = 8$. Distribution-level results refine this: MatrixRes leads in the degree- and density-driven BA, ER, and regular-degree families, whose labels separate related structural regimes rather than unrelated categories, while community-driven SBM is less favourable. Taken together, matrix-style reuse is conditionally effective rather than universal, and is most useful when the task requires separating several related structural regimes under finer discrimination.

D. Limitations

Several factors bound these findings. The benchmark uses a fixed graph-level pooling and classification head, so alternative readouts may interact with residual topology. The compared operators are canonical message-passing backbones rather than recent graph transformers or local-global hybrids, which is a deliberate control choice but means the study is not a state-of-the-art comparison. The sensitivity scan is fold-0 only, and the candidate reruns cover MatrixResGated on PROTEINS and DD alone. ENZYMES has low absolute accuracy and high variability. The mechanism analysis is correlational: branch diversity, cosine similarity, CKA, and gradient norms support a coherent explanation but do not prove causality. Finally, the benchmark intentionally avoids task-specific chemical or biological feature engineering, so absolute accuracies should not be read as optimized state-of-the-art molecular prediction

results, and the synthetic families stress-test residual topology rather than substitute for broader real multi-class benchmarks. Future work should extend the comparison to larger datasets, richer readouts, and learned residual-neighborhood policies that replace the hand-specified stencils.

VI. CONCLUSION

This paper studied residual connectivity for graph classification as a structured branch-by-layer design problem. Across six datasets and four message-passing operators, residual topology matters, but its value depends on the dataset, operator, and branch budget. MatrixRes is the most frequent winner with the best mean rank in the full benchmark, yet the per-dataset GCNConv winners remain mixed, and ENZYMES has low absolute accuracy and should be read cautiously. Branch-count ablations show that adding branches helps only within a limited operating region: additional branches raise representational diversity but also residual traffic, and they can preserve redundant branch similarity or weaken optimization. The synthetic multi-class benchmark supports a conditional reading—matrix-style reuse is unnecessary for the easiest two- and three-class settings but becomes more competitive from four classes onward, where MatrixResGated attains the best average and normalized accuracy and MatrixRes degrades least from two to eight classes. The practical implication is that residual topology, branch count, operator choice, and residual control should be evaluated jointly, and that candidate settings should be rerun across folds before a sensitivity result is treated as a performance claim.

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DATA AVAILABILITY

The code and experiment records supporting this study are available from the corresponding author on reasonable request.

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